

Use of SNAP to convert Planet Tanager hyperspectral imagery into GIS-ready formats

Emil A. Cherrington, Ph.D.

Lab for Applied Science / Earth System Science Center

University of Alabama in Huntsville

Scenario: You are a GIS user
and would like to open and
view the Tanager data in
ArcGIS or *QGIS*.

Objective: Use SNAP
software to convert Tanager
HDF5 data to GeoTIFF
format (*which can be read by
ArcGIS or QGIS*)

Methods

1. Select + download Tanager data from the open STAC
 - basic_sr_hdf5
2. [Install +] Open SNAP v.13
3. Open HDF file (File -> Open Product)
4. Reproject data (Raster -> Geometric -> Reproject)
5. Resample data (Raster -> Geometric -> Resample)
6. Export data (File -> Export -> GeoTIFF / BigTIFF)

Tanager Core Imagery

https://www.planet.com/data/stac/browser/tanager-core-imagery/catalog.json?.language=en

planet

Tanager Core Imagery

in Planet Labs - Open Data

Catalog Source Share Language: English

Description

Tanager Core Imagery Sample Products - Basic (georeferenced, unprojected) and orthorectified (georeferenced, projected) core imagery data products from Tanager-1 hyperspectral satellite. The data included in this Tanager STAC Catalogue (the "Licensed Material") is made available under the Creative Commons CC-BY 4.0 license (<https://creativecommons.org/licenses/by/4.0/>). Use the following attribution for any redistribution of the Licensed Materials, "Tanager STAC Data, available at www.planet.com/data/stac © [YEAR] Planet Labs PBC. All Rights Reserved", and preface with "Adapted from..." for any Adapted Material. Where [YEAR] means the year in which the data was captured by the Tanager satellite, as indicated for each data collection.

Catalogs

Filter catalogs by title, description or keywords

Select keywords

Agriculture Basic (georeferenced, unprojected) and ortho (georeferenced, projected) core imagery data products from Tanager-1 for Agriculture applications 2/23/2025, 4:55:38 PM UTC - 9/29/2025, 8:59:22 AM UTC	Fire Basic (georeferenced, unprojected) and ortho (georeferenced, projected) core imagery data products from Tanager-1 for Fire applications 12/15/2024, 6:59:16 PM UTC - 9/20/2025, 7:32:07 PM UTC	ROCX 2025 Basic (georeferenced, unprojected) and ortho (georeferenced, projected) core imagery data products from Tanager-1 for ROCX 2025 applications 7/4/2025, 4:52:04 PM UTC - 10/29/2025, 4:55:01 PM UTC
Coastal and Water Bodies Basic (georeferenced, unprojected) and ortho (georeferenced, projected) core imagery data products from Tanager-1 for Coastal and Water Bodies... 2/23/2025, 4:55:46 PM UTC - 10/29/2025, 4:54:55 PM UTC	GHG Plumes Basic (georeferenced, unprojected) and ortho (georeferenced, projected) core imagery data products from Tanager-1 for GHG Plumes applications 10/4/2024, 8:19:21 AM UTC - 4/23/2025, 1:40:26 PM UTC	Snow and Ice Basic (georeferenced, unprojected) and ortho (georeferenced, projected) core imagery data products from Tanager-1 for Snow and Ice applications 4/11/2025, 4:34:21 PM UTC - 6/6/2025, 6:12:48 PM UTC
Energy and Mining Basic (georeferenced, unprojected) and ortho (georeferenced, projected) core imagery data products from Tanager-1 for Energy and Mining applications 9/25/2024, 6:55:04 PM UTC - 9/11/2025, 7:15:47 PM UTC	Natural Lands Basic (georeferenced, unprojected) and ortho (georeferenced, projected) core imagery data products from Tanager-1 for Natural Lands applications 9/25/2024, 6:55:04 PM UTC - 9/29/2025, 8:59:22 AM UTC	Urban Basic (georeferenced, unprojected) and ortho (georeferenced, projected) core imagery data products from Tanager-1 for Urban applications 10/25/2024, 6:50:51 PM UTC - 10/29/2025, 4:55:01 PM UTC

<https://www.planet.com/data/stac/browser/tanager-core-imagery/catalog.json>

Coastal and Water Bodies

https://www.planet.com/data/stac/browser/tanager-core-imagery/coastal-water-bodies/collection.json

planet

Coastal and Water Bodies

Catalog

Source

Share

Language: English

in Planet Labs - Open Data

Up

Browse

Description

Basic (georeferenced, unprojected) and ortho (georeferenced, projected) core imagery data products from Tanager-1 for Coastal and Water Bodies applications

tanager

hyperspectral

coastal/water bodies

License

CC-BY-4.0

Temporal Extent

2/23/2025, 4:55:46 PM UTC - 10/29/2025, 4:54:55 PM UTC

+

-



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Asset

> product_specifications

GRAPHIC

METADATA

SPECIFICATIONS

PDF

Providers

> Planet Labs PBC

PRODUCER

PROCESSOR

> Google Cloud Platform

HOST

Items 43

Ascending

Descending



TanagerScene 20250223_165546_32_4001 Core Imagery

Hdf5

COG

All data products from Tanager-1 for 20250223_165546_32_4001 near Usulután, 3401, El Salvador

2/23/2025, 4:55:46 PM UTC



TanagerScene 20250225_153627_16_4001 Core Imagery

Hdf5

COG

All data products from Tanager-1 for 20250225_153627_16_4001 near Puerto Maraven, Parroquia Cardón, Municipio Cariubana, Falcon State, 4154, Venezuel...

2/25/2025, 3:36:27 PM UTC



TanagerScene 20250824_171857_84_4001

https://www.planet.com/data/stac/browser/tanager-core-imagery/coastal-water-bodies/20250824_1...

School

planet

TanagerScene 20250824_171857_84_4001 Core Imagery

Catalog

Source

Share

Language: English

in Planet Labs - Open Data

Up

Collection

Browse

+

-

Reserva E. Bama de Chetumal Santuario del Mana

Calderitas

Sergio Buitrón Casas

Chetumal

Javier Rojo Gómez

Shoastern insersion management Area

Corozal

Corozal Bay Wildlife Sanctuary

Map

Thumbnails

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Description

All data products from Tanager-1 for 20250824_171857_84_4001 near Corozal Town, Corozal, Belize

Collection

Coastal and Water Bodies

Basic (georeferenced, unprojected) and ortho (georeferenced, projected) core imagery data products from Tanager-1 for Coastal and Water Bodies applications

2/23/2025, 4:55:46 PM UTC - 10/29/2025, 4:54:55 PM UTC

Metadata

General

Cloud Percent	6
Collection Mode	standard_sensitivity
Constellation	Tanager
Time of Data	8/24/2025, 5:18:57 PM UTC
GSD	32.86 m
Instruments	4001
License	CC-BY-4.0
Light Haze Percent	0
Location Description	Corozal Town, Corozal, Belize
Notes	n/a
Platform	Planet

Assets

> basic_beta_udm

METADATA

CLOUD

DATA-MASK

COG

> basic_radiance_hdf5

DATA

METADATA

CLOUD

DATA-MASK

HDF5

> basic_sr_hdf5

REFLECTANCE

DATA

METADATA

CLOUD

DATA-MASK

HDF5

> geolocation_array

METADATA

COG

> ortho_beta_udm

METADATA

CLOUD

DATA-MASK

COG

> ortho_radiance_hdf5

DATA

METADATA

CLOUD

DATA-MASK

HDF5

> ortho_sr_hdf5

REFLECTANCE

DATA

METADATA

CLOUD

DATA-MASK

HDF5

> ortho_visual

SHOWN

VISUAL

DATA

COG

> thumbnail

THUMBNAIL

PNG

Caveat: We focus on the `basic_sr_hdf5` product because we were unable to successfully export the `ortho_sr_hdf5` product w/ SNAP. That may change in the future.



[Catalog](#)
[Source](#)
[Share](#)
Language: English

in Planet Labs - Open Data

↑ Up

Collection

 Browse

Assets

> basic_beta_udm

METADATA CLOUD DATA-MASK COG

- basic radiance hdf5

DATA METADATA CLOUD DATA-MASK HDF5

```
> basic_sr_hdf5
```

REFLECTANCE DATA METADATA CLOUD DATA-MASK HDF5

- > geolocation array

METADATA COG

```
> ortho_beta_udm
```

METADATA CLOUD DATA-MASK COG

```
> ortho_radiance_hdf5
```

DATA METADATA CLOUD DATA-MASK HDF5

```
> ortho_sr_hdf5
```

REFLECTANCE DATA METADATA CLOUD DATA-MASK HDF5

> ortho visual

✓ SHOWN VISUAL DATA COG

> thumbnail

THUMBNAIL PNG

Description

All data products from Tanager-1 for 20250824_171857_84_4001 near Corozal Town, Corozal, Belize

Collection

Coastal and Water Bodies

Basic (georeferenced, unprojected) and ortho (georeferenced, projected) core imagery data products from Tanager-1 for Coastal and Water Bodies applications

2/23/2025, 4:55:46 PM UTC - 10/29/2025, 4:54:55 PM UTC

Metadata

General

Cloud Percent	6
Collection Mode	standard_sensitivity
Constellation	Tanager
Time of Data	8/24/2025, 5:18:57 PM UTC
GSD	32.86 m
Instruments	4001
License	CC-BY-4.0
Light Haze Percent	0
Location Description	Corozal Town, Corozal, Belize
Notes	<i>n/a</i>
Platform	Planet

TanagerScene 20250824_17185

https://www.planet.com/data/stac/browser/tanager-core-imagery/coastal-water-bodies/20250...

planet

ortho_sr_hdf5

DATA

METADATA

CLOUD

DATA-MASK

HDF5

ortho_sr_hdf5

REFLECTANCE

DATA

METADATA

CLOUD

DATA-MASK

HDF5

Hdf5

Download

Copy URL

Surface reflectance and other associated datasets as an orthorectified (georeferenced and projected) HDFEOS formatted HDF5 file. Note that nearest neighbor resampling is used in orthorectification and reflectance values are stored directly as floats in a 0-1 range so no scaling/offset is needed.

Metadata

Electro-Optical

Spectral Bands

Name	Wavelength	FWHM
surface_reflectance_B000	0.37644 μm	0.00539 μm
surface_reflectance_B001	0.38141 μm	0.00542 μm
surface_reflectance_B002	0.38638 μm	0.00545 μm
surface_reflectance_B003	0.39135 μm	0.00548 μm
surface_reflectance_B004	0.39632 μm	0.00552 μm
surface_reflectance_B005	0.40129 μm	0.00555 μm
surface_reflectance_B006	0.40626 μm	0.00558 μm
surface_reflectance_B007	0.41123 μm	0.0056 μm

Projection

Code	EPSG:7030
Image Dimensions	671 × 763

Raster Imagery

Bands

License

CC-BY-4.0

Light Haze Percent

0

Location Description

Corozal Town, Corozal, Belize

Notes

n/a

Platform

Planet

Plume Provider Ids

n/a

Quality Category

standard

View Geometry

Viewing Azimuth

315.8 °

Off-Nadir Angle

1.3 °

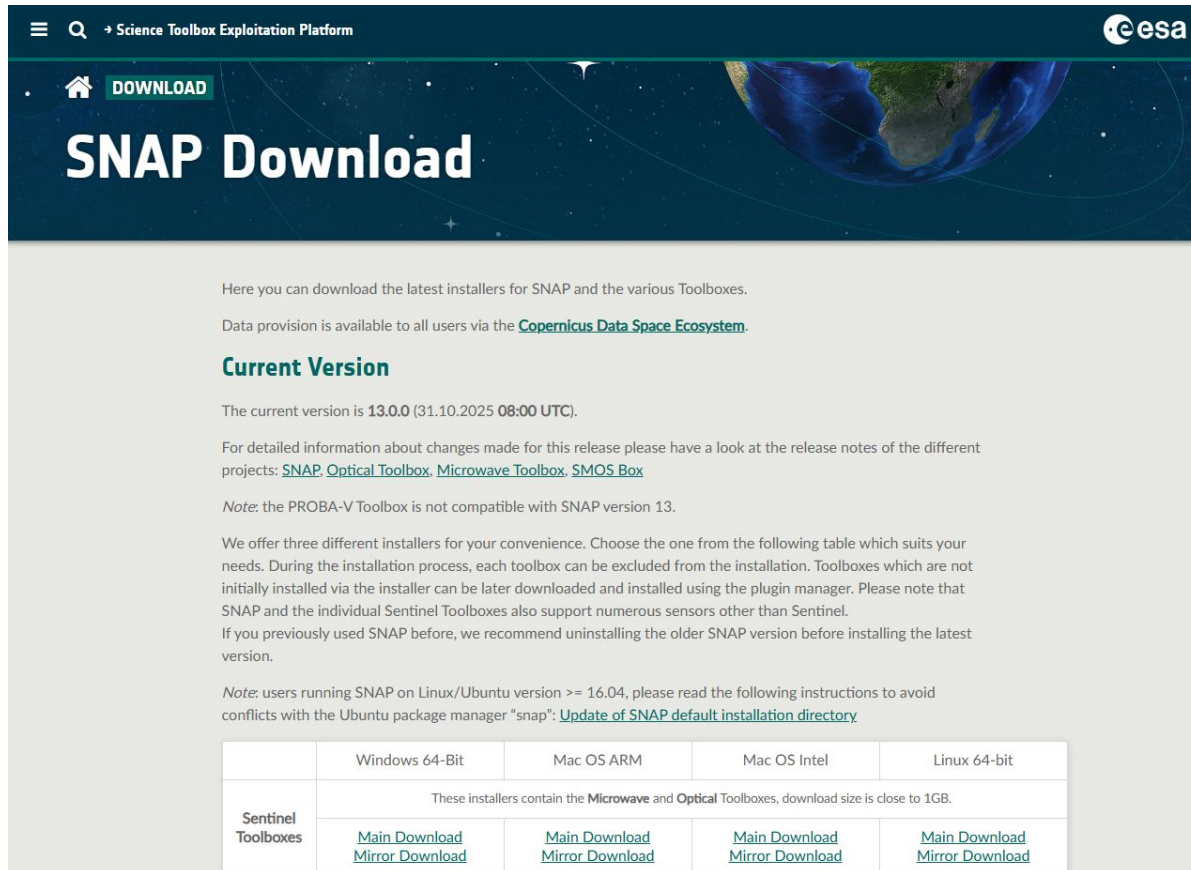
Sun Azimuth

129.1 °

Sun Elevation

78.3 °

1. Download ESA's SNAP (v.13)



The screenshot shows the 'Science Toolbox Exploitation Platform' header with the ESA logo. Below is a 'DOWNLOAD' button and the title 'SNAP Download'. The main content area contains text about downloading installers, a link to the 'Copernicus Data Space Ecosystem', and the 'Current Version' section. The current version is 13.0.0 (31.10.2025 08:00 UTC). It provides links to release notes for SNAP, Optical Toolbox, Microwave Toolbox, and SMOS Box. A note states that the PROBA-V Toolbox is not compatible with SNAP version 13. It offers three different installers for Windows 64-Bit, Mac OS ARM, Mac OS Intel, and Linux 64-bit. A table lists download links for Sentinel Toolboxes, including Main Download and Mirror Download for each platform.

Here you can download the latest installers for SNAP and the various Toolboxes.

Data provision is available to all users via the [Copernicus Data Space Ecosystem](#).

Current Version

The current version is **13.0.0** (31.10.2025 08:00 UTC).

For detailed information about changes made for this release please have a look at the release notes of the different projects: [SNAP](#), [Optical Toolbox](#), [Microwave Toolbox](#), [SMOS Box](#)

Note: the PROBA-V Toolbox is not compatible with SNAP version 13.

We offer three different installers for your convenience. Choose the one from the following table which suits your needs. During the installation process, each toolbox can be excluded from the installation. Toolboxes which are not initially installed via the installer can be later downloaded and installed using the plugin manager. Please note that SNAP and the individual Sentinel Toolboxes also support numerous sensors other than Sentinel. If you previously used SNAP before, we recommend uninstalling the older SNAP version before installing the latest version.

Note: users running SNAP on Linux/Ubuntu version ≥ 16.04 , please read the following instructions to avoid conflicts with the Ubuntu package manager "snap": [Update of SNAP default installation directory](#)

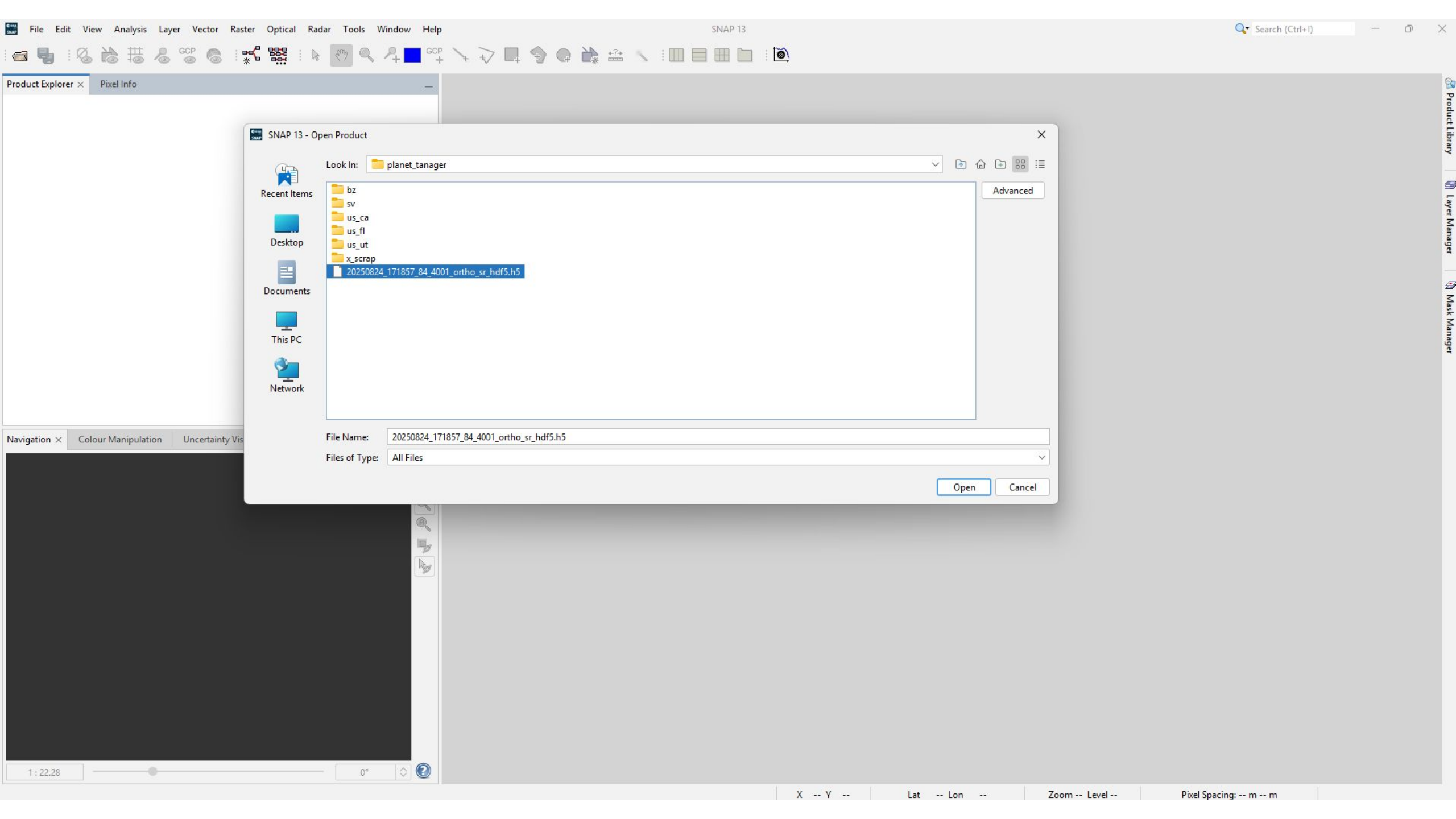
	Windows 64-Bit	Mac OS ARM	Mac OS Intel	Linux 64-bit
Sentinel Toolboxes	These installers contain the Microwave and Optical Toolboxes, download size is close to 1GB.			
	Main Download Mirror Download	Main Download Mirror Download	Main Download Mirror Download	Main Download Mirror Download

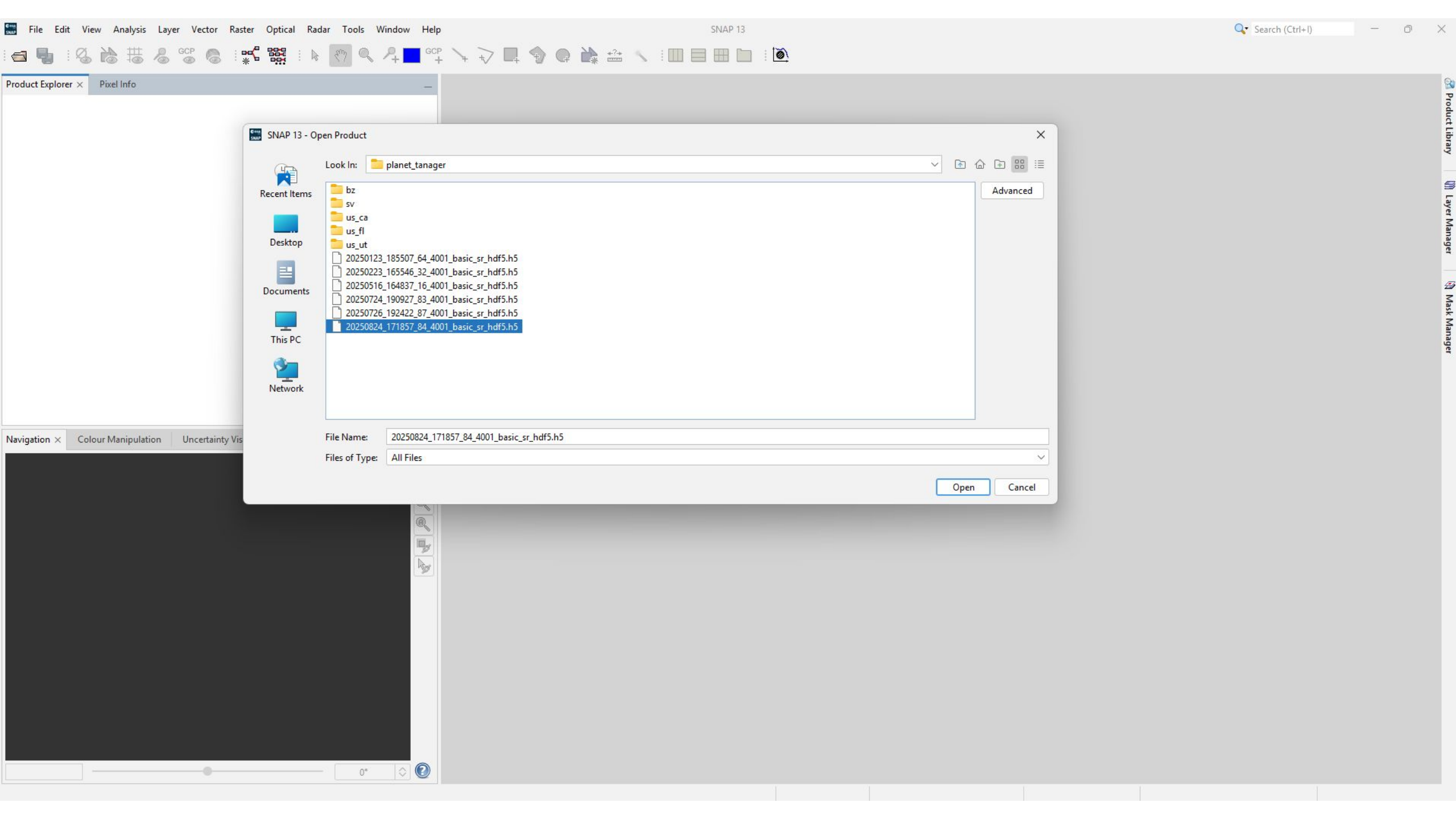
2. Install, then open SNAP

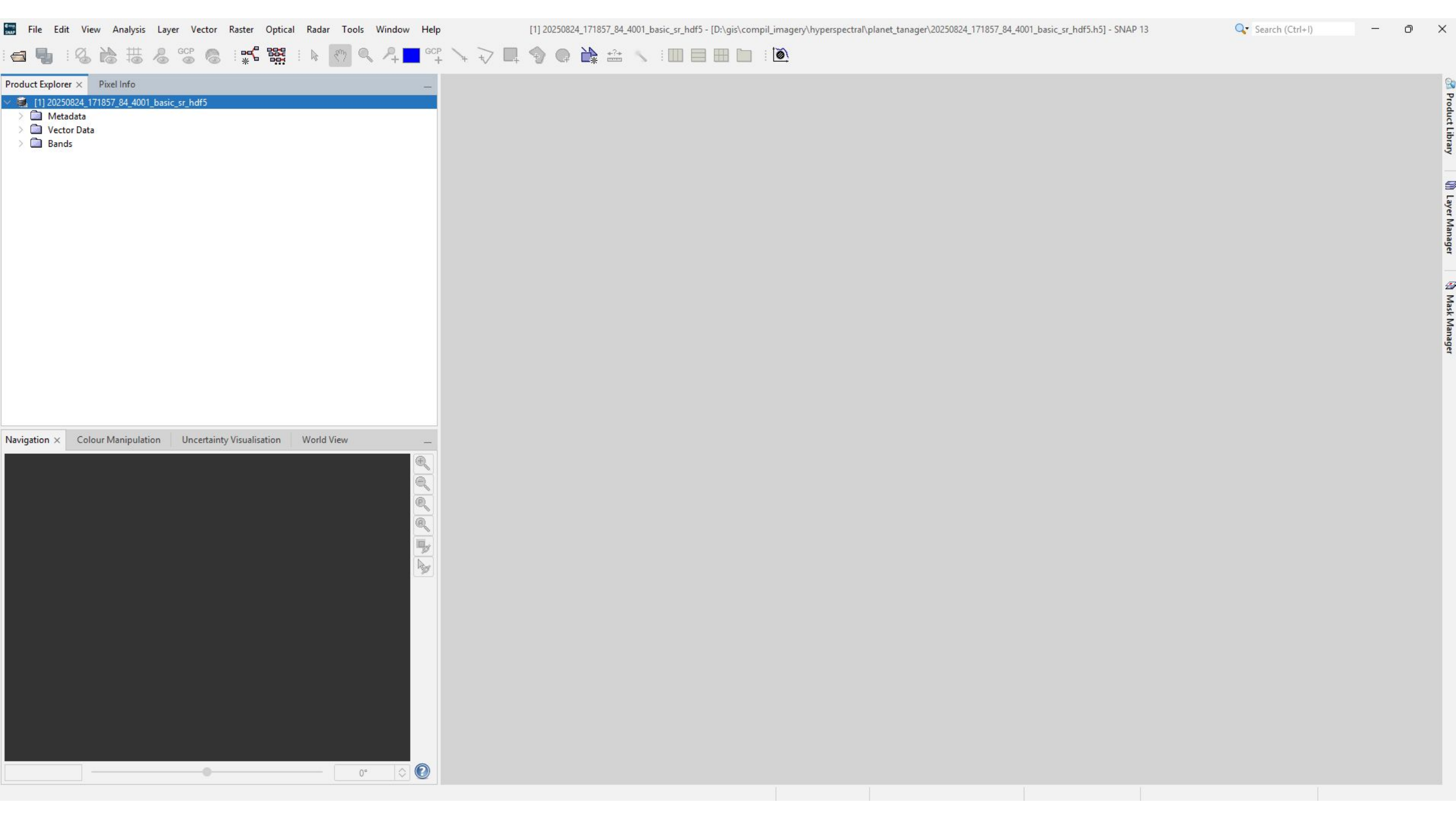


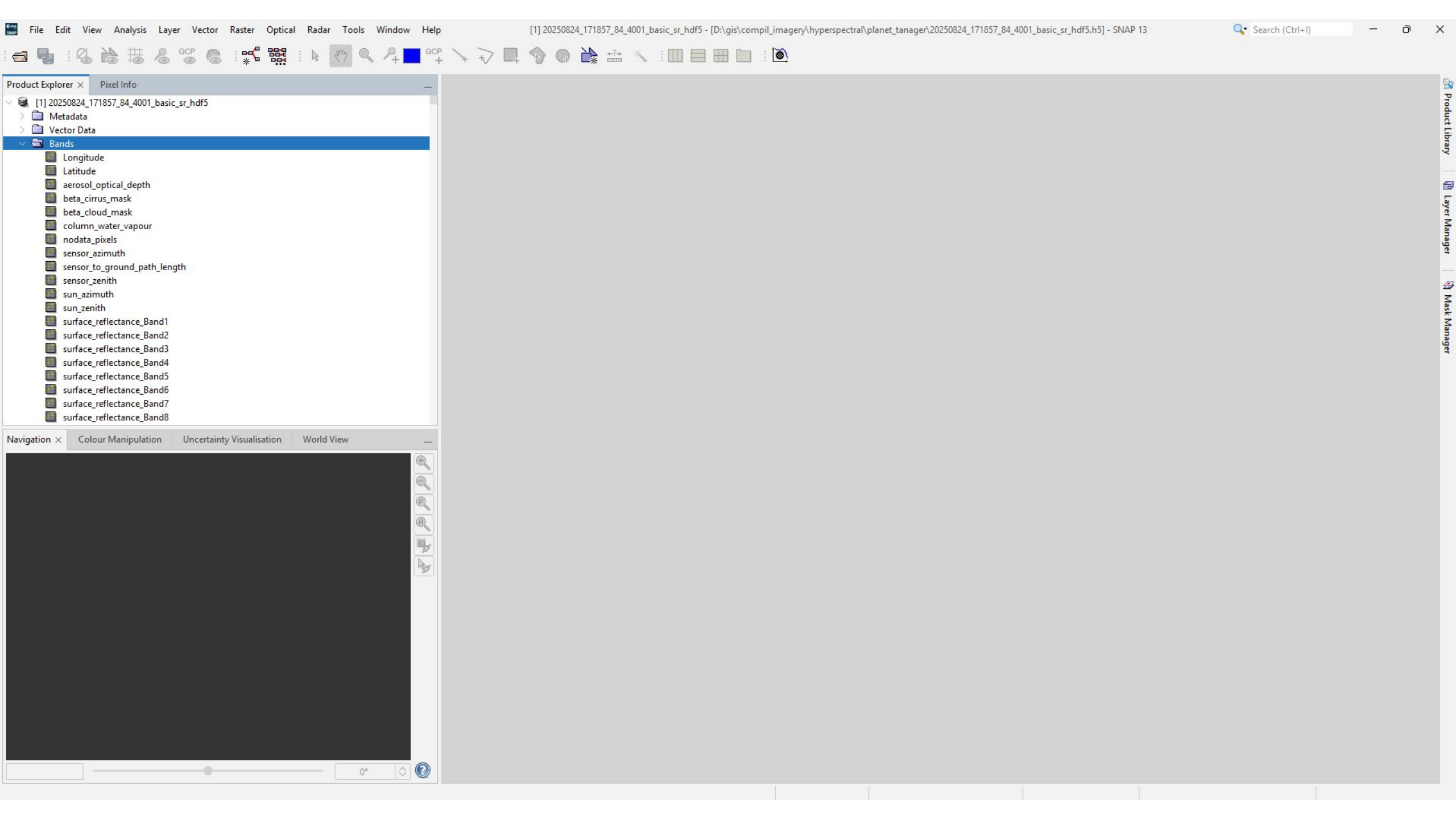
<https://step.esa.int/main/download/snap-download/>

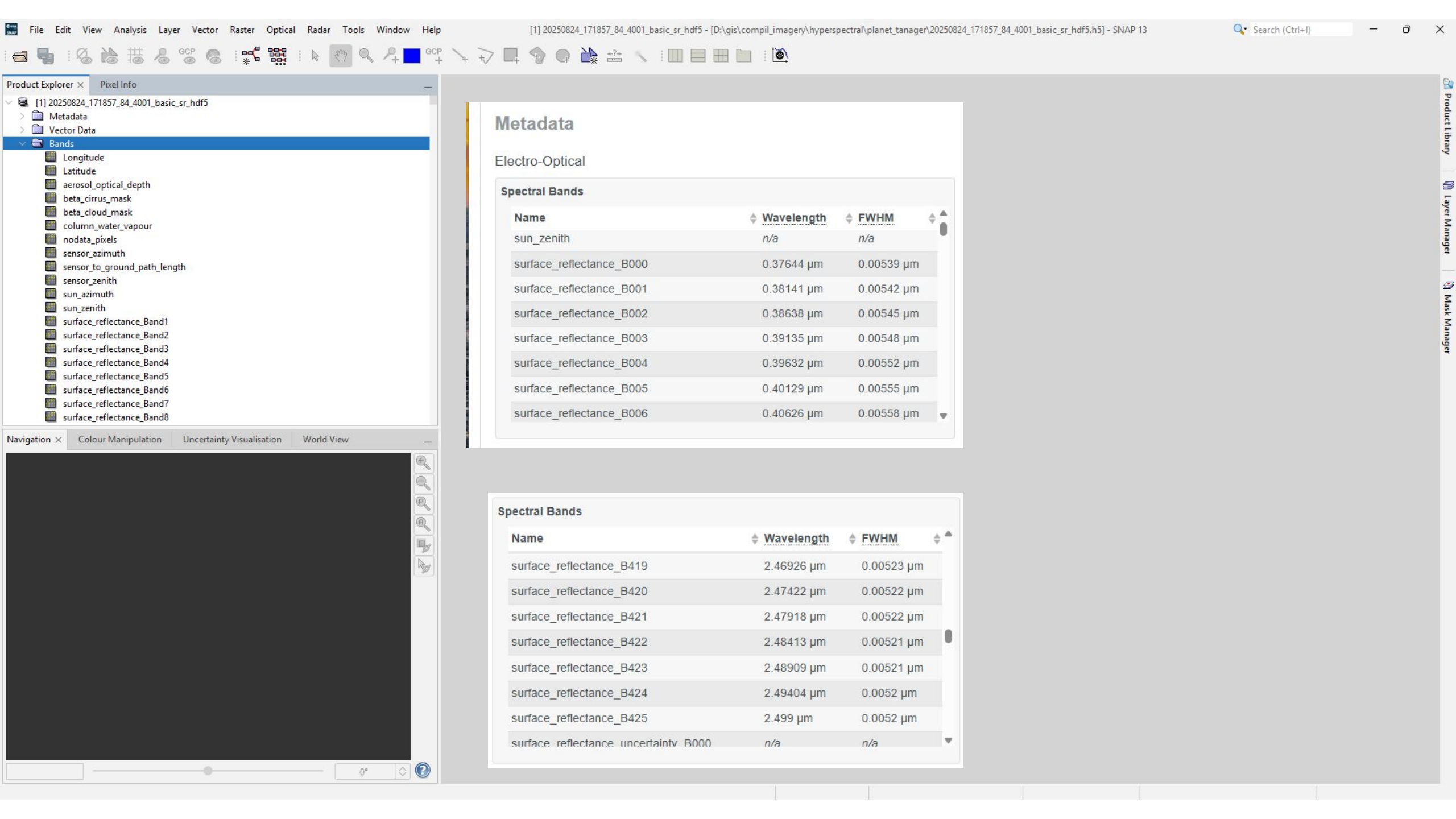
Convert `basic_sr_hdf5` to
GeoTIFF











Metadata

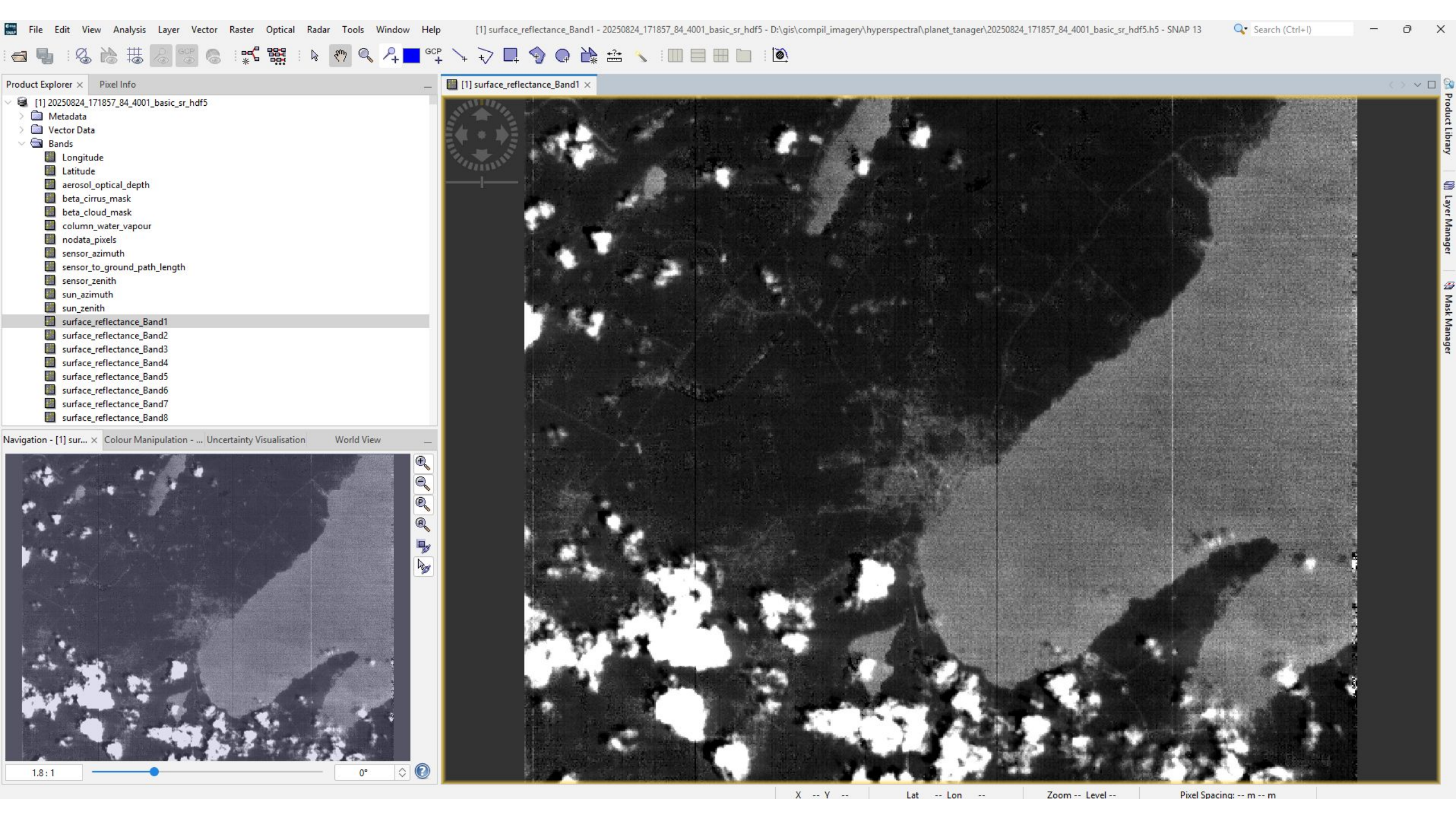
Electro-Optical

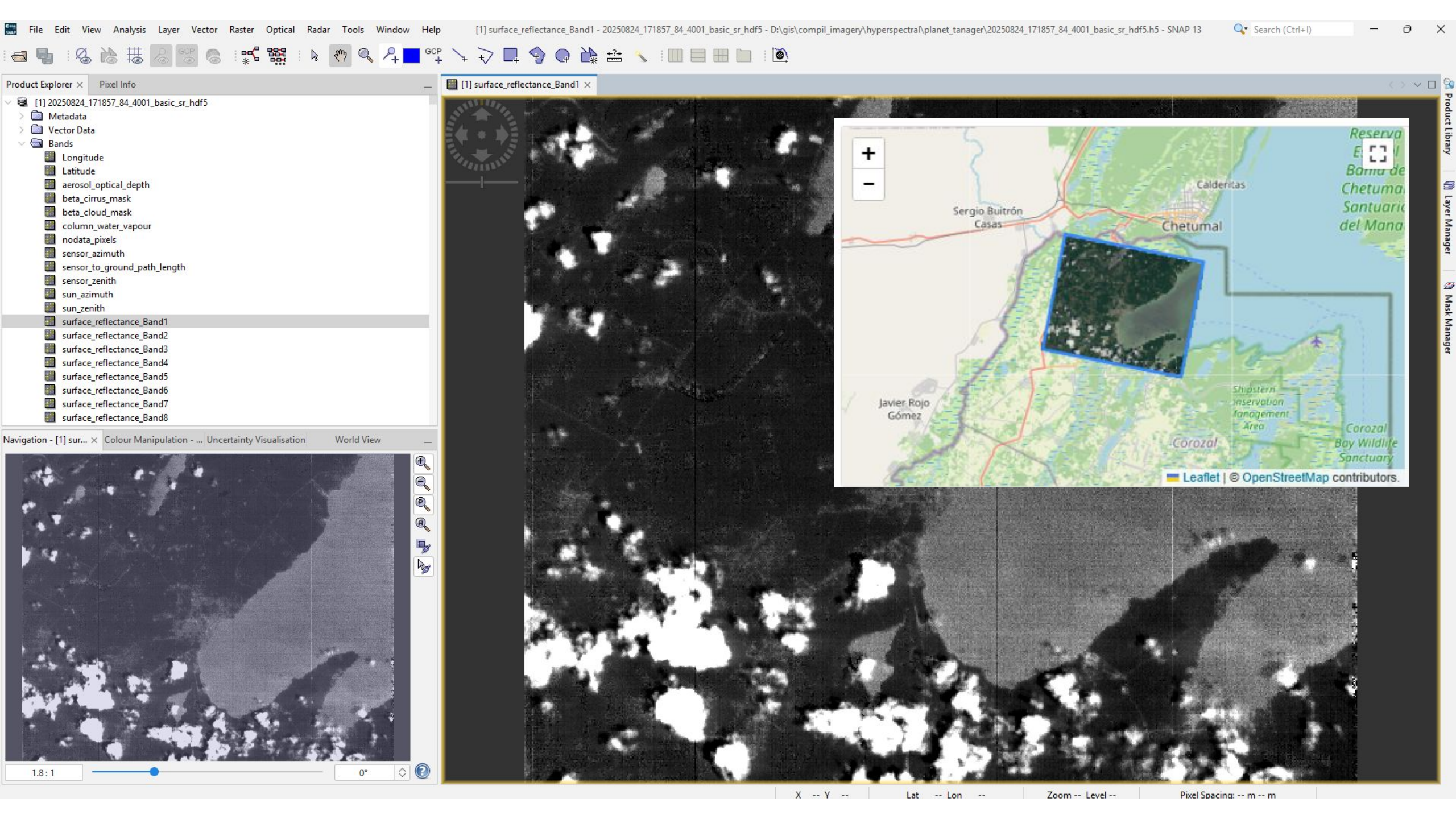
Spectral Bands

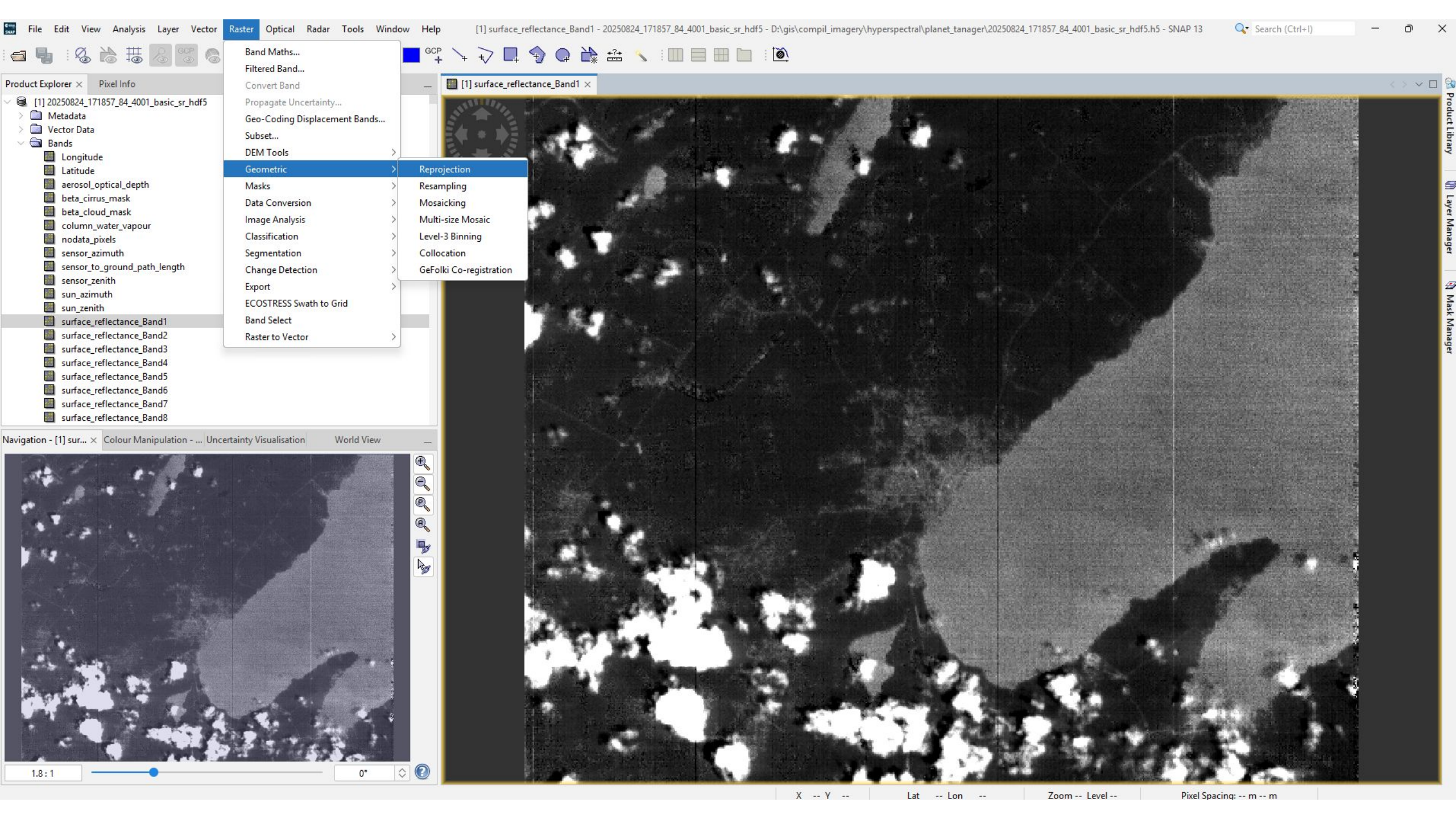
Name	Wavelength	FWHM
sun_zenith	n/a	n/a
surface_reflectance_B000	0.37644 μm	0.00539 μm
surface_reflectance_B001	0.38141 μm	0.00542 μm
surface_reflectance_B002	0.38638 μm	0.00545 μm
surface_reflectance_B003	0.39135 μm	0.00548 μm
surface_reflectance_B004	0.39632 μm	0.00552 μm
surface_reflectance_B005	0.40129 μm	0.00555 μm
surface_reflectance_B006	0.40626 μm	0.00558 μm

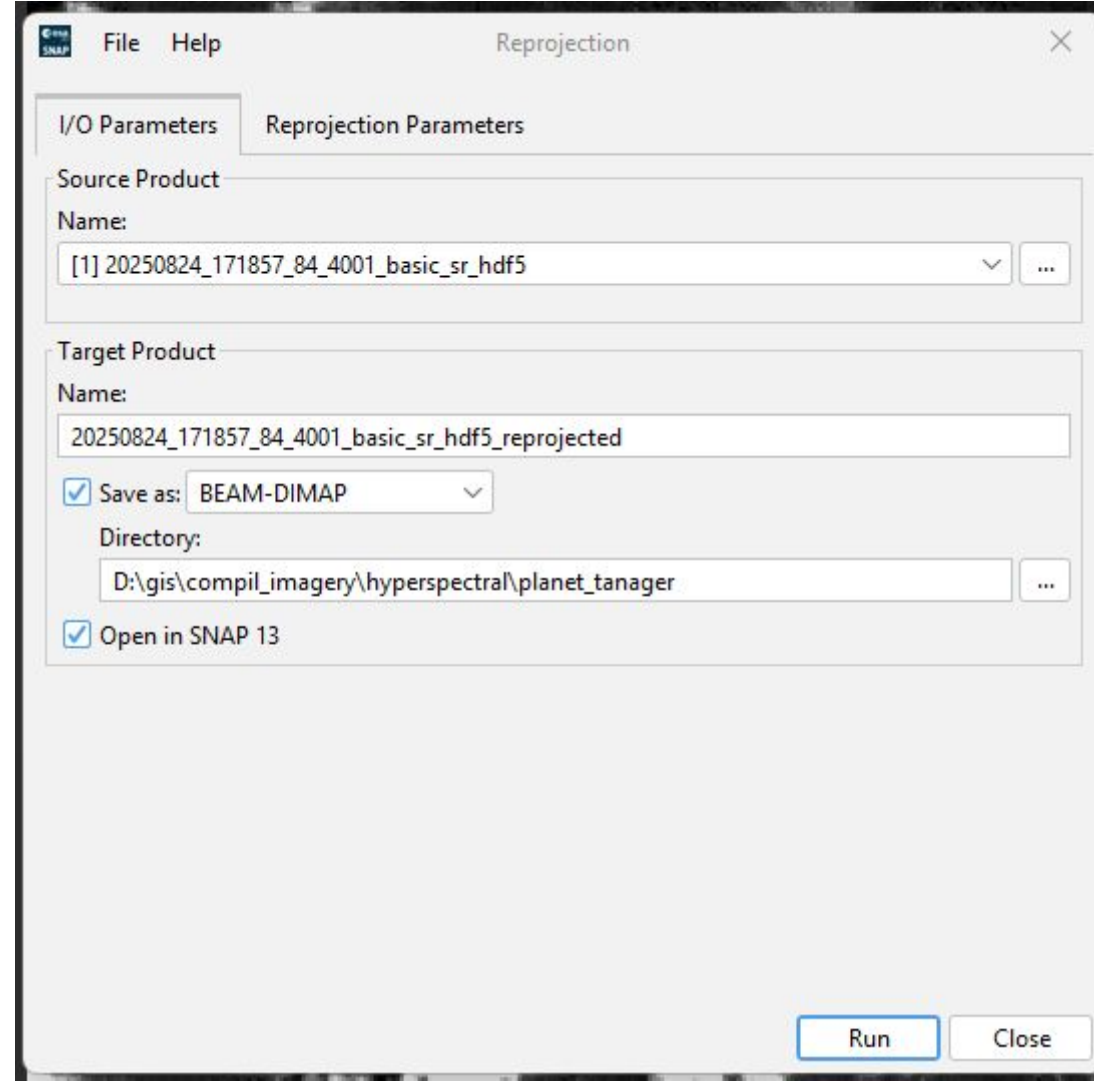
Spectral Bands

Name	Wavelength	FWHM
surface_reflectance_B419	2.46926 μm	0.00523 μm
surface_reflectance_B420	2.47422 μm	0.00522 μm
surface_reflectance_B421	2.47918 μm	0.00522 μm
surface_reflectance_B422	2.48413 μm	0.00521 μm
surface_reflectance_B423	2.48909 μm	0.00521 μm
surface_reflectance_B424	2.49404 μm	0.0052 μm
surface_reflectance_B425	2.499 μm	0.0052 μm
surface_reflectance_uncertainty_B000	n/a	n/a









File Help Reprojection

I/O Parameters Reprojection Parameters

Coordinate Reference System (CRS)

☒ Custom CRS

Geodetic datum: World Geodetic System 1984

Projection: Geographic Lat/Lon (WGS 84)

Projection Parameters...

☐ Predefined CRS Select...

☐ Use CRS of ...

Output Settings

☒ Preserve resolution ☒ Reproject tie-point grids

Output Parameters...

No-data value: NaN

☐ Add delta lat/lon bands Resampling method: Nearest

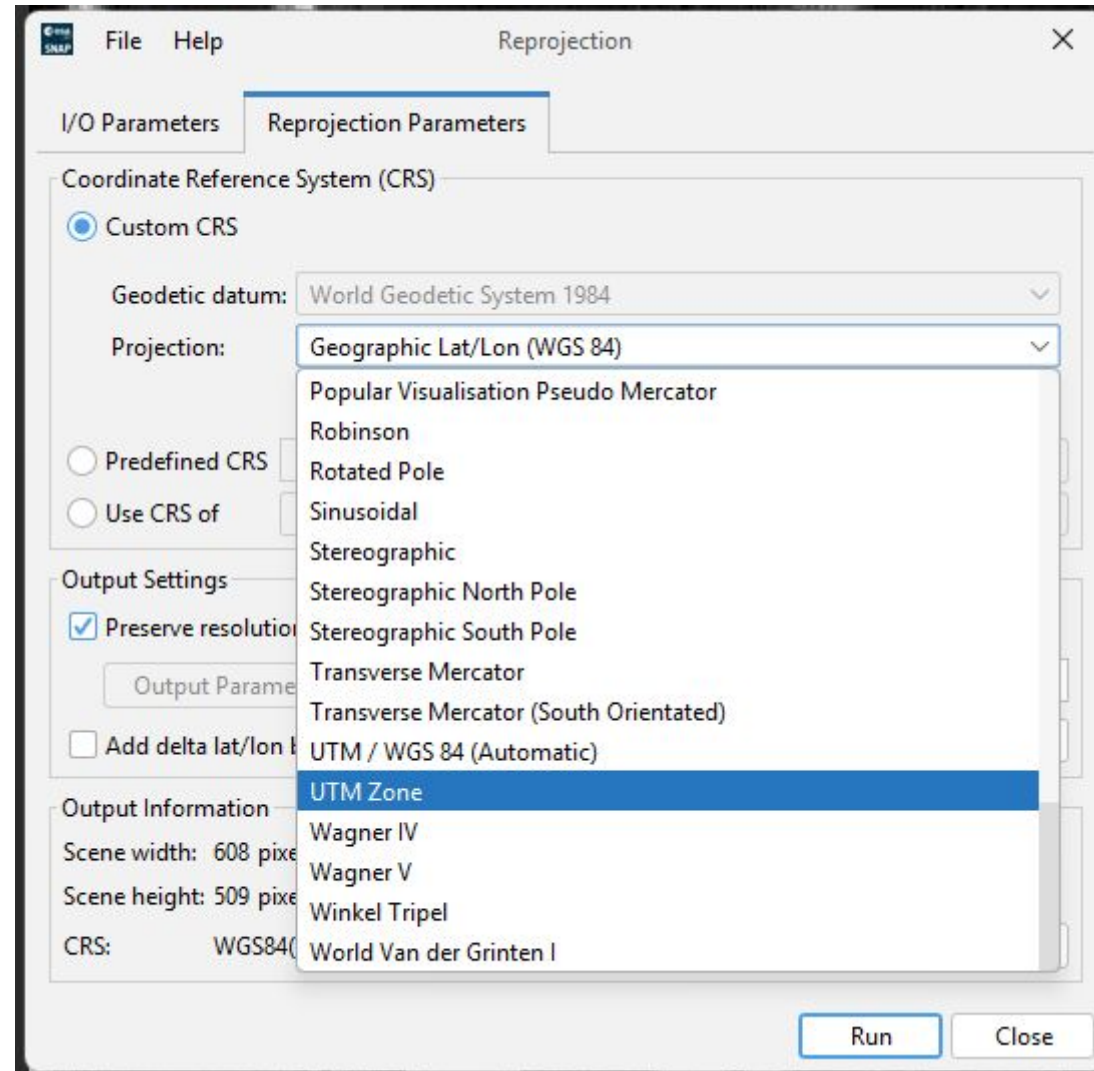
Output Information

Scene width: 608 pixel Center longitude: 88°23'16" W

Scene height: 509 pixel Center latitude: 18°24'02" N

CRS: WGS84(DD) Show WKT

Run Close



FileHelp

Reprojection

I/O Parameters

Reprojection Parameters

Coordinate Reference System (CRS)

☒ Custom CRS

Geodetic datum:World Geodetic System 1984

Projection:UTM Zone

Projection Parameters...

☐ Predefined CRS

Select...

☐ Use CRS of

Output Settings

☒ Preserve resolution

☒ Reproject tie-point grids

Output Parameters...

No-data value:NaN

☐ Add delta lat/lon bands

Resampling method:Nearest

Output Information

Scene width: 608 pixel

Center longitude: 88°23'16" W

Scene height: 581 pixel

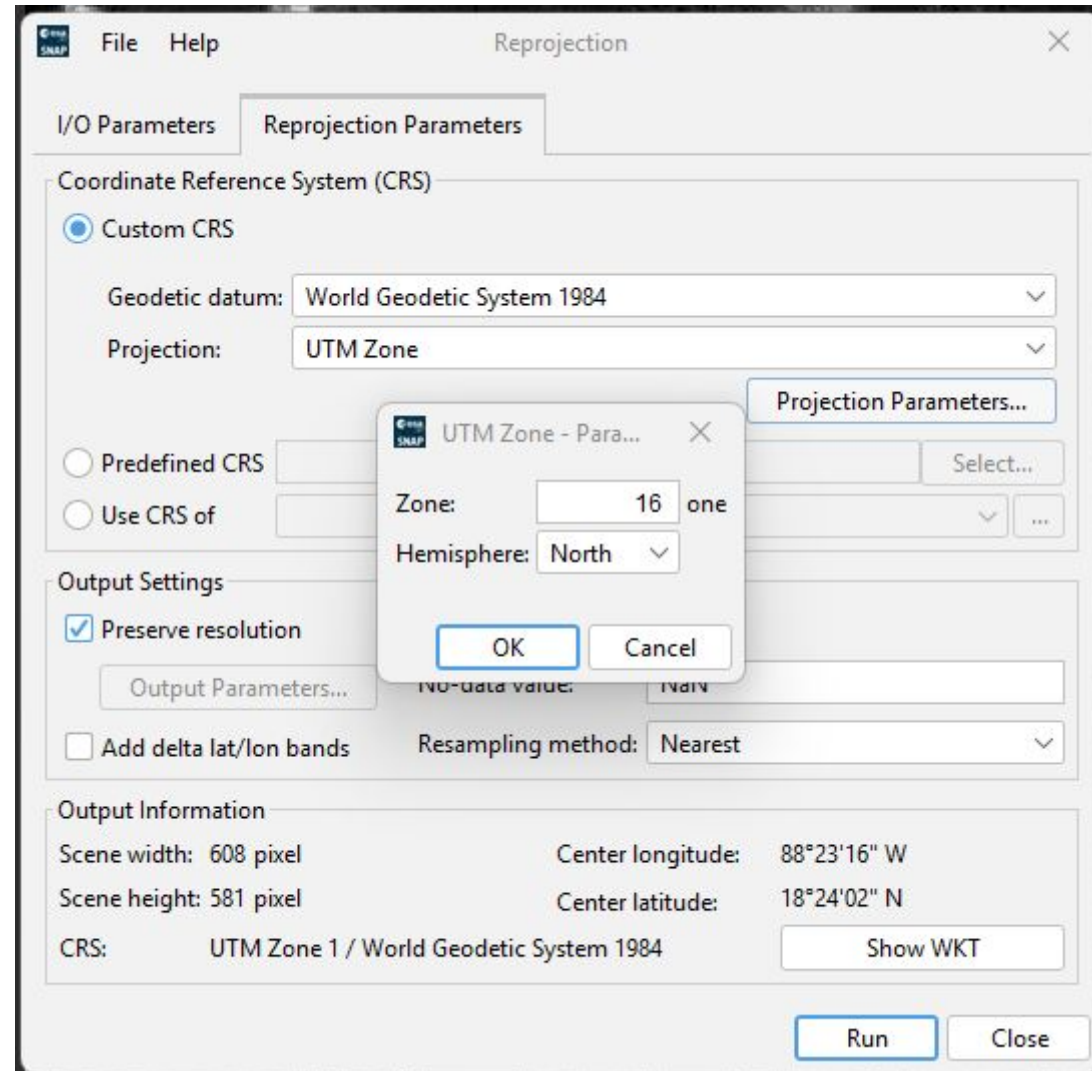
Center latitude: 18°24'02" N

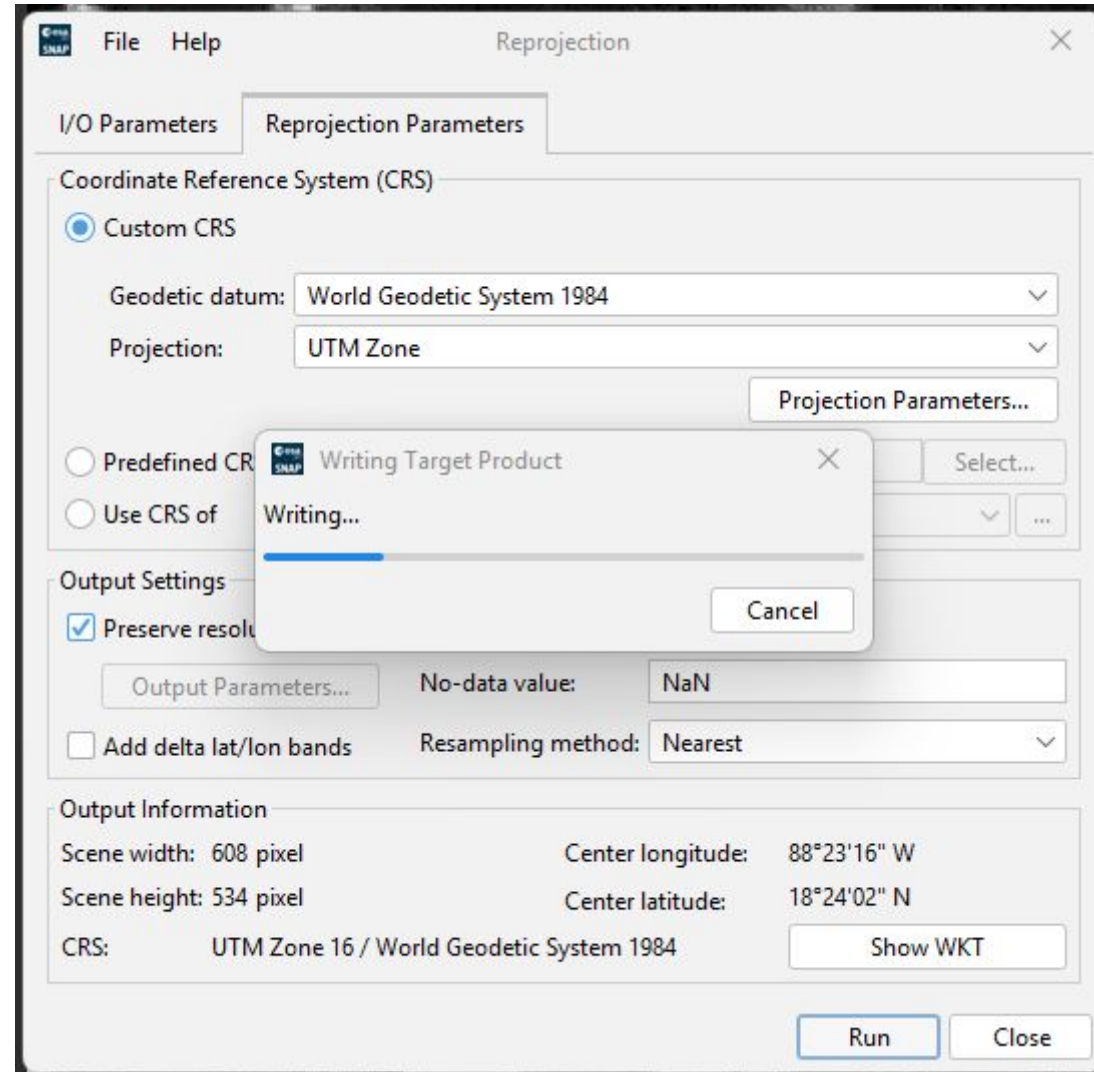
CRS:UTM Zone 1 / World Geodetic System 1984

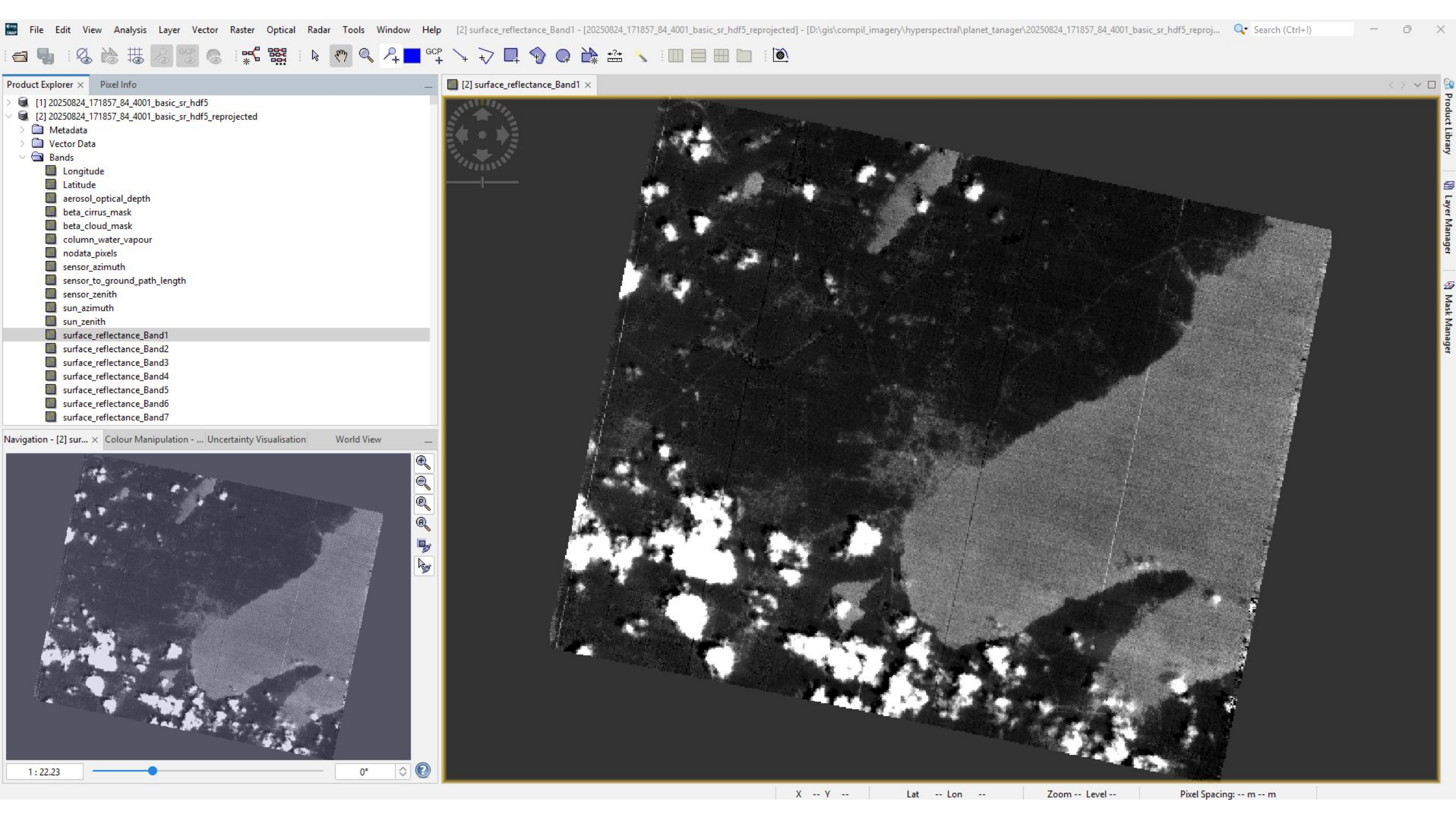
Show WKT

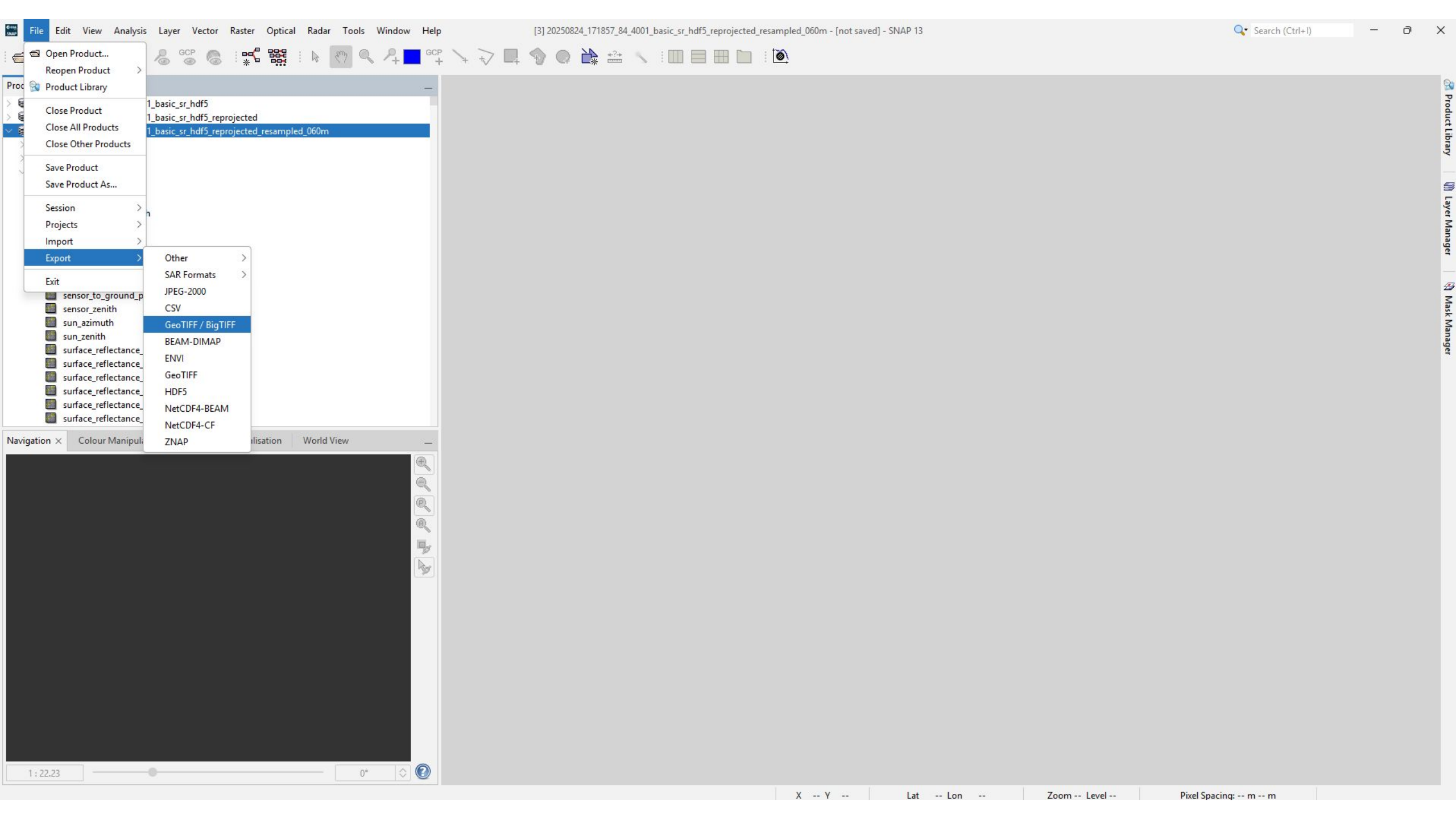
Run

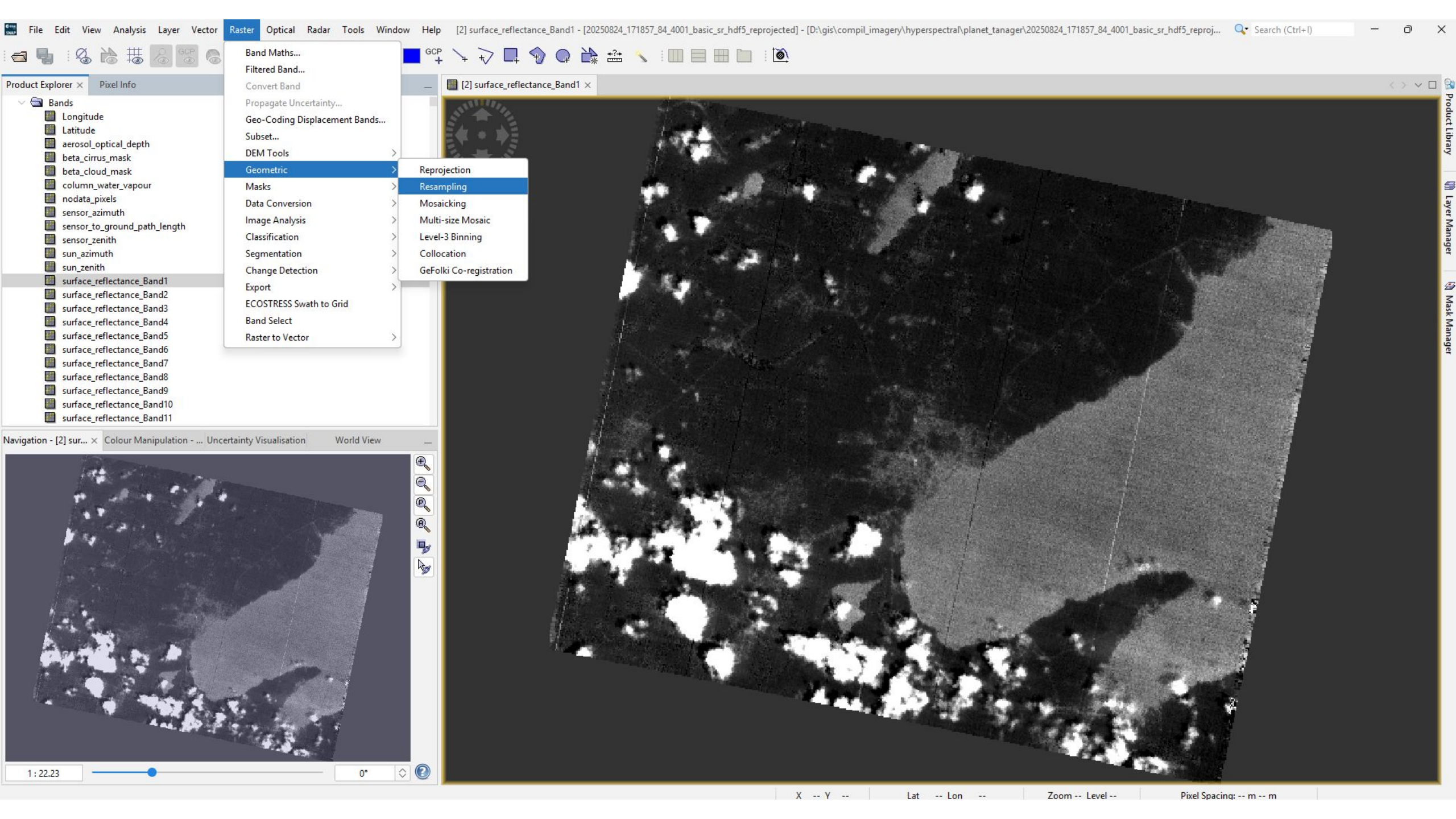
Close











File Help Resampling

I/O Parameters Resampling Parameters

Source Product

Name:

[2] 20250824_171857_84_4001_basic_sr_hdf5_reprojected

Target Product

Name:

20250824_171857_84_4001_basic_sr_hdf5_reprojected_resampled

☐ Save as: BEAM-DIMAP

Directory:

D:\gis\compil_imagery\hyperspectral\planet_tanager

☒ Open in SNAP 13

Run Close

File

Help

Resampling

I/O Parameters

Resampling Parameters

Define size of resampled product

☐ By reference band from source product:

Longitude

Resulting target width: 608

Resulting target height: 534

☐ By target width and height:

Target width: 608

Target height: 534

Width / height ratio: 1.13858

☒ By pixel resolution (in m):

60

Resulting target width: 381

Resulting target height: 335

Define resampling algorithm

Upsampling method:

Nearest

Downsampling method:

First

Flag downsampling method:

First

☐ Advanced Method Definition by Band

☒ Resample on pyramid levels (for faster imaging)

Run

Close

File

Help

Resampling

✕

I/O Parameters

Resampling Parameters

Define size of resampled product

☐ By reference band from source product:

Longitude

Resulting target width: 608

Resulting target height: 534

☐ By target width and height:

Target width: 608

Target height: 534

Width / height ratio: 1.13858

☒ By pixel resolution (in m):

60

Resulting target width: 381

Resulting target height: 335

Define resampling algorithm

Upsampling method:

Nearest

Downsampling method:

First

First

Min

Max

Mean

Median

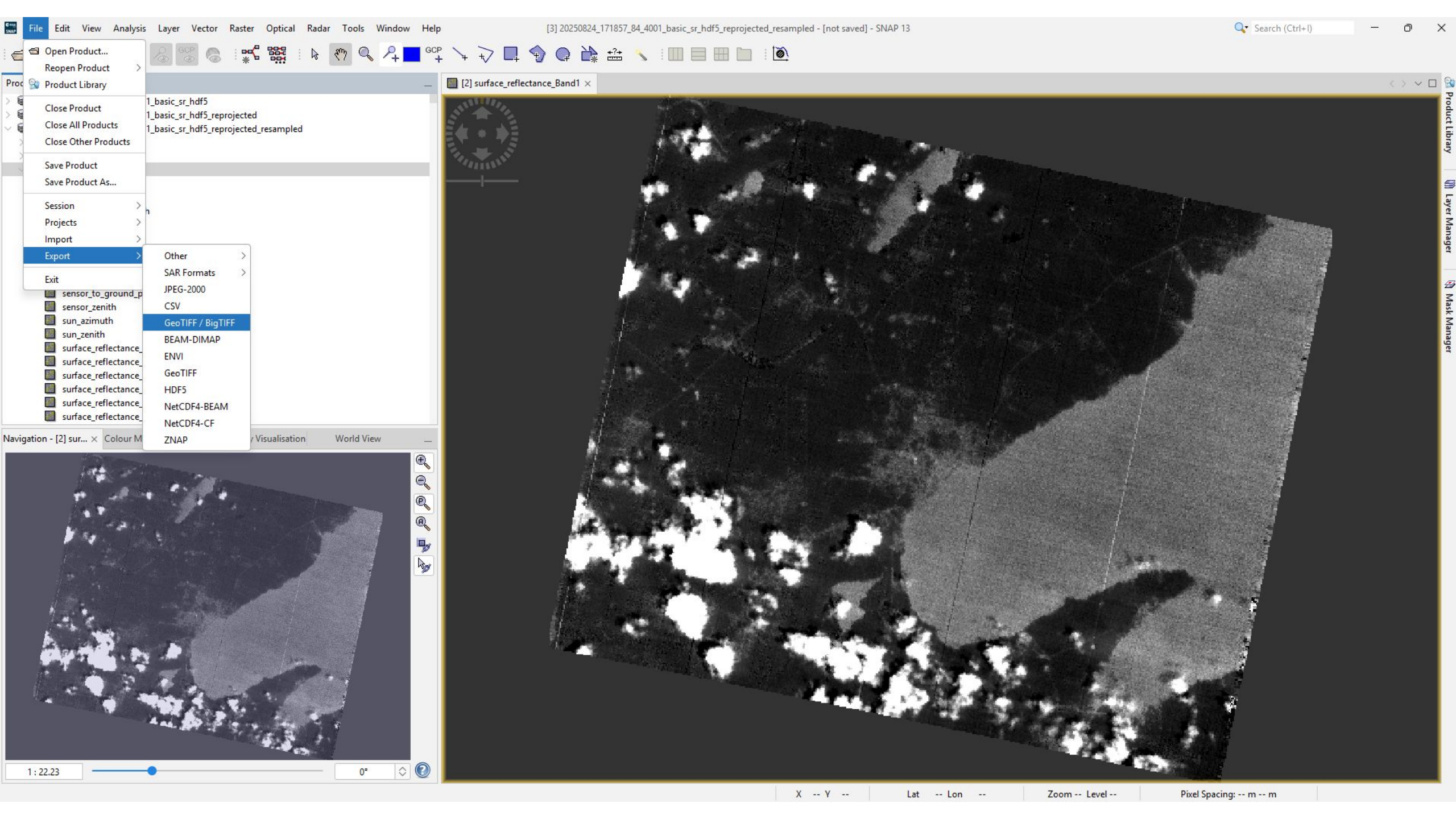
Flag downsampling method:

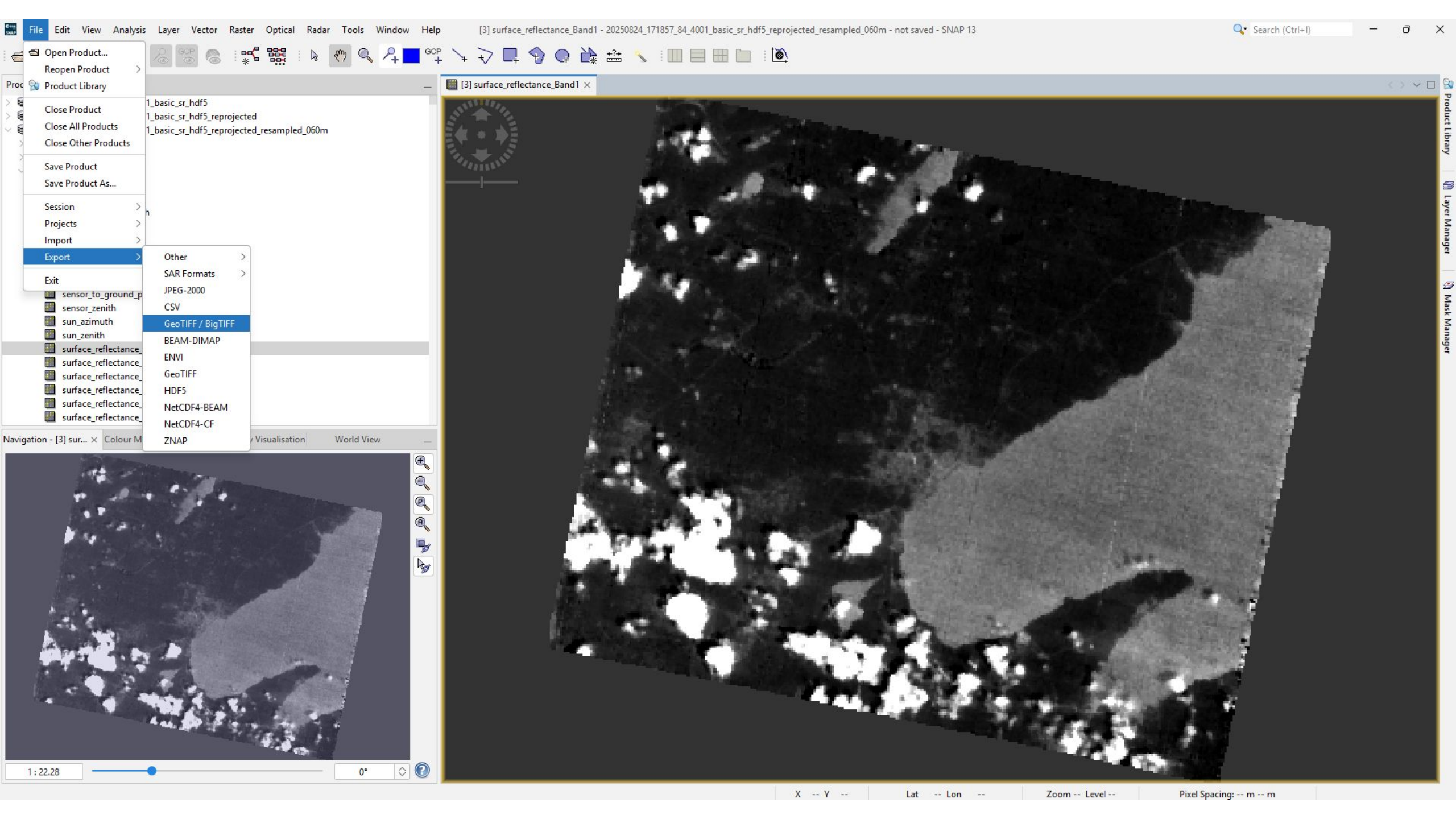
☐ Advanced Method Definition by Band

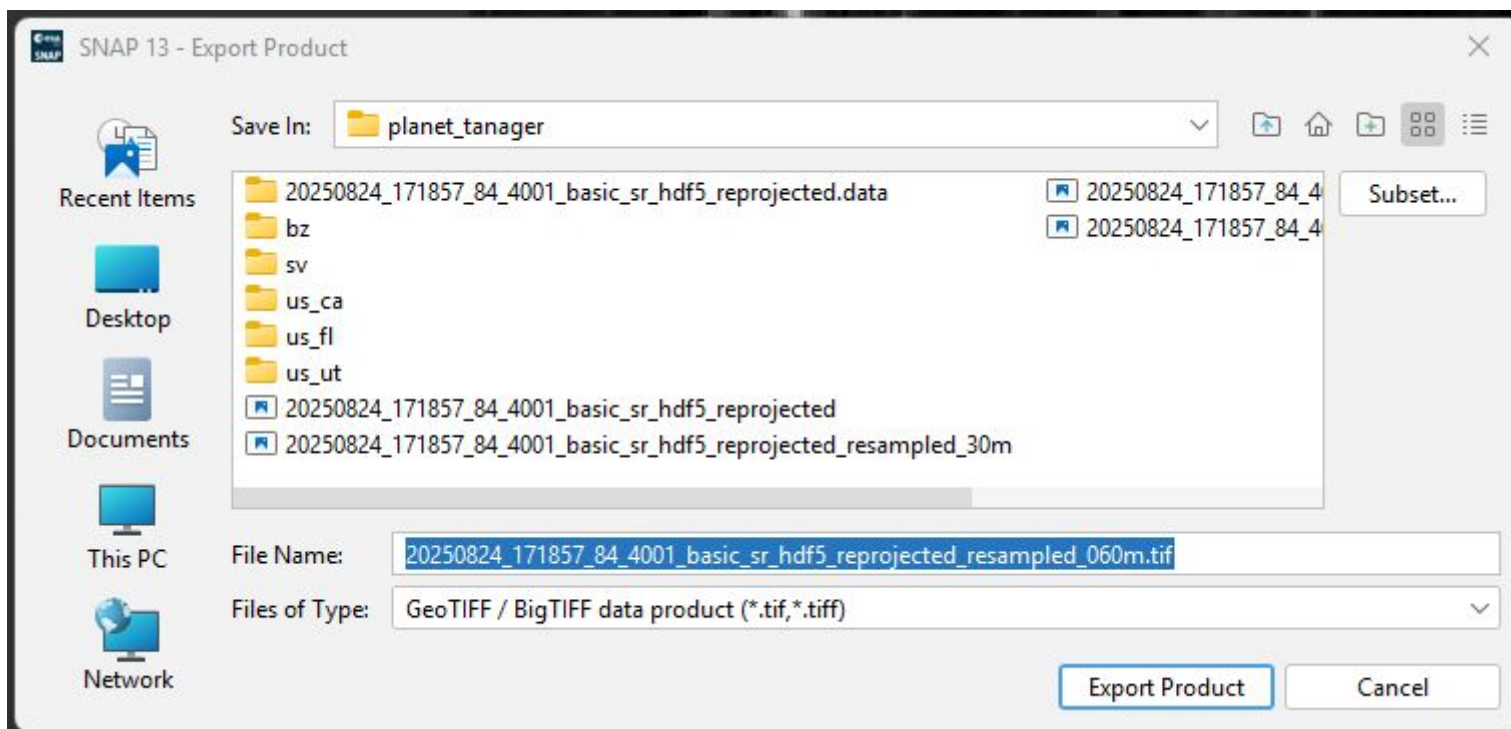
☒ Resample on pyramid levels (for faster imaging)

Run

Close







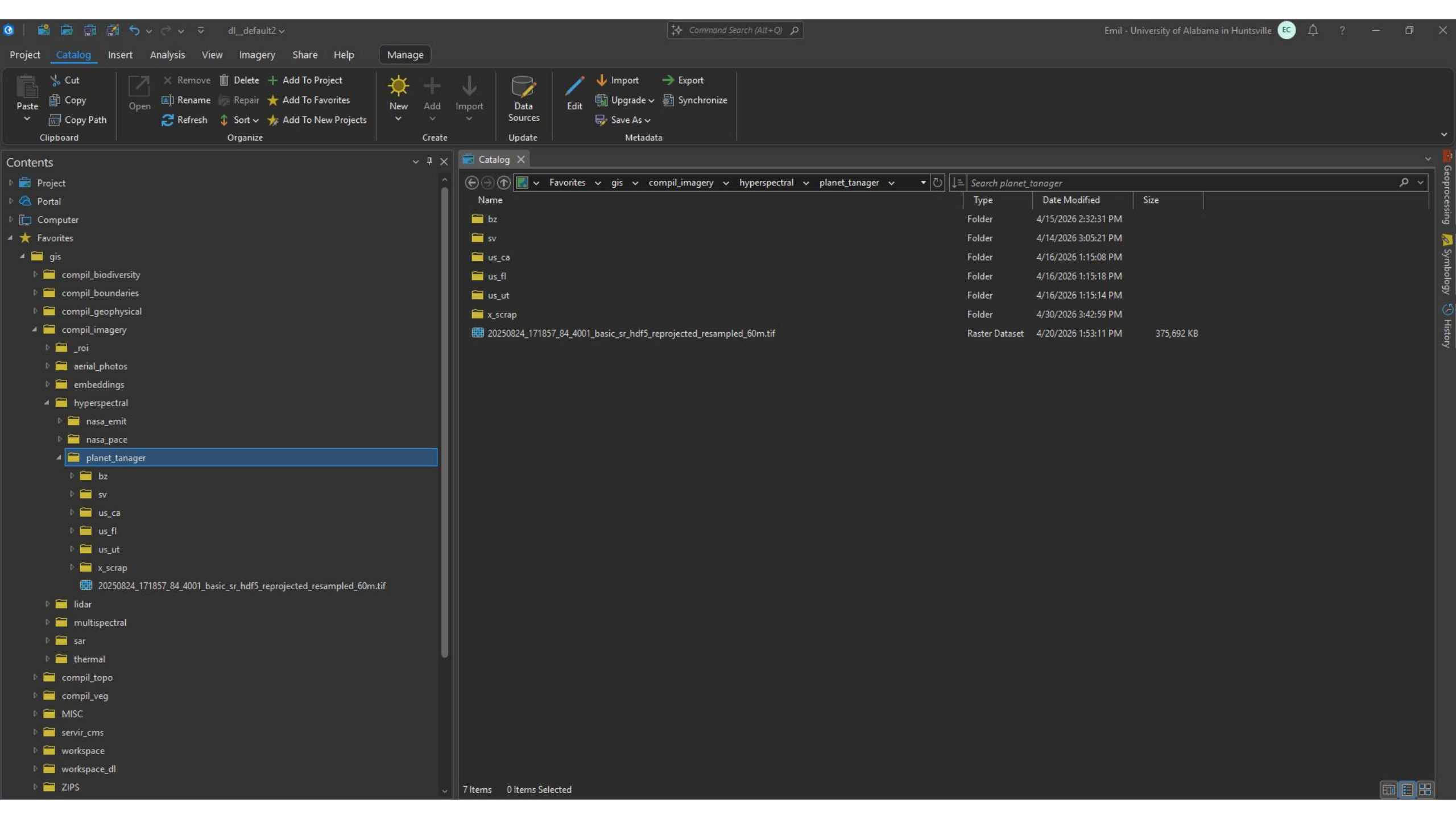


ArcGIS® Pro

Initializing



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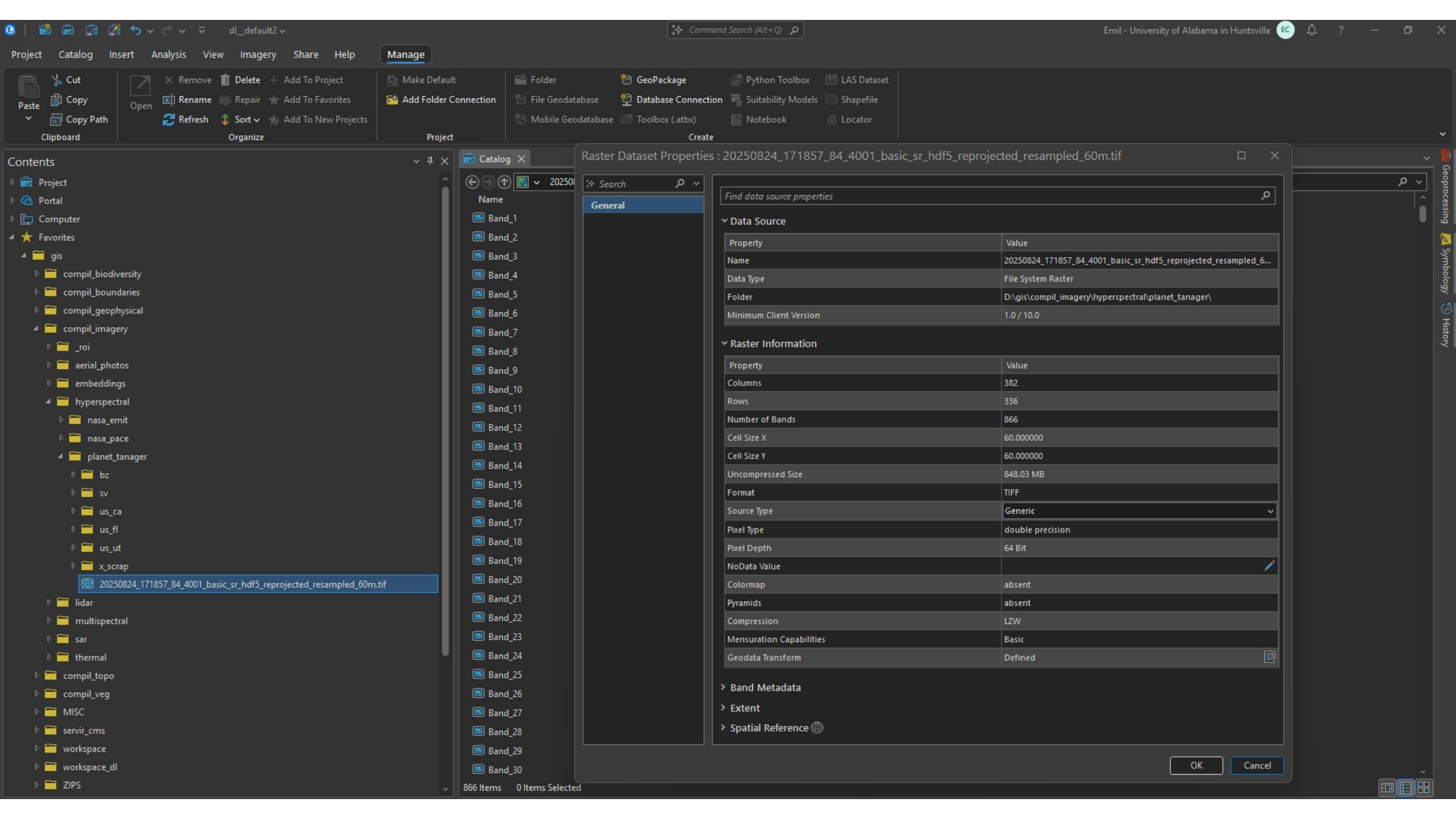
- Contents
- Project
 - Portal
 - Computer
 - Favorites
 - gis
 - compil_biodiversity
 - compil_boundaries
 - compil_geophysical
 - compil_imagery
 - _roi
 - aerial_photos
 - embeddings
 - hyperspectral
 - nasa_emit
 - nasa_pace
 - planet_tanager
 - bz
 - sv
 - us_ca
 - us_fl
 - us_ut
 - x_scrap
 - 20250824_171857_84_4001_basic_sr_hdf5_reprojected_resampled_60m.tif
 - lidar
 - multispectral
 - sar
 - thermal
 - compil_topo
 - compil_veg
 - MISC
 - servir_cms
 - workspace
 - workspace_dl
 - ZIPS

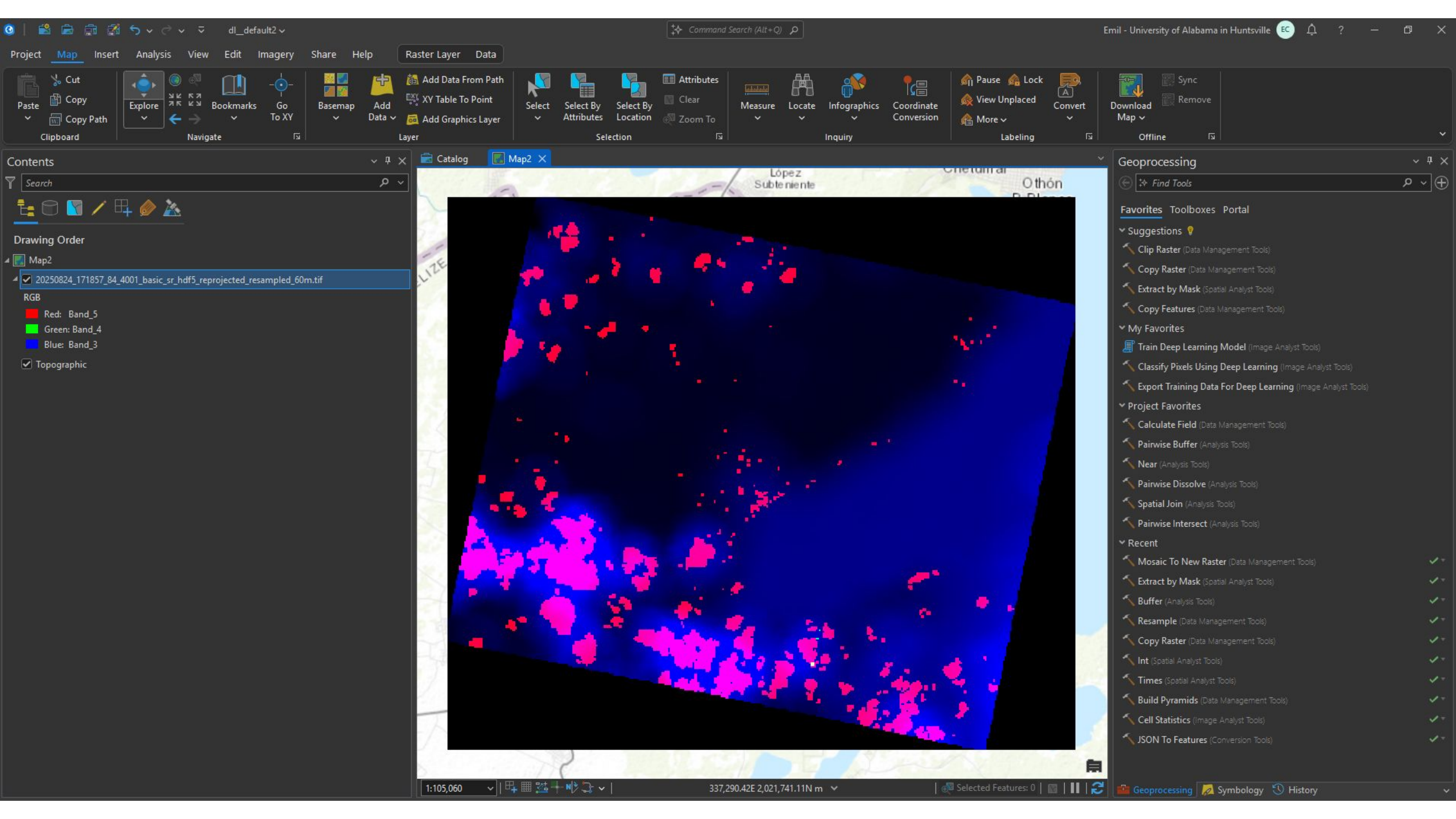
Catalog

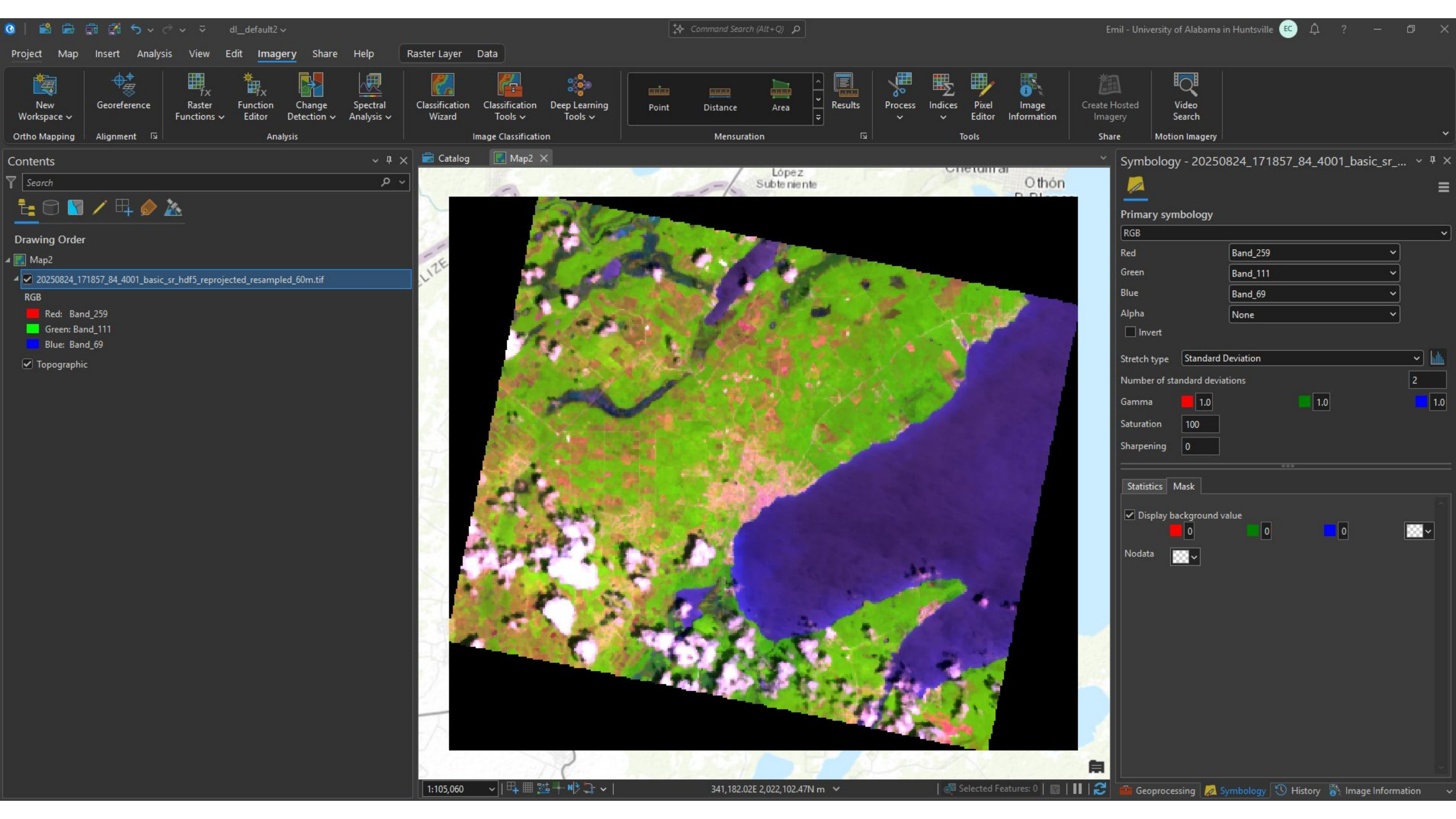
Search planet_tanager

Name	Type	Date Modified	Size
bz	Folder	4/15/2026 2:32:31 PM	
sv	Folder	4/14/2026 3:05:21 PM	
us_ca	Folder	4/16/2026 1:15:08 PM	
us_fl	Folder	4/16/2026 1:15:18 PM	
us_ut	Folder	4/16/2026 1:15:14 PM	
x_scrap	Folder	4/30/2026 3:42:59 PM	
20250824_171857_84_4001_basic_sr_hdf5_reprojected_resampled_60m.tif	Raster Dataset	4/20/2026 1:53:11 PM	375,692 KB

7 Items 0 Items Selected







Questions?

emil.cherrington@uah.edu